

CS 767 Advanced Machine Learning Homework 4

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1. **Scikit-learn tutorial.** The TA has prepared a detailed tutorial explaining the basics of Data Analysis with Scikit-learn using Linear Regression as an example. The tutorial also discusses cross validation. Go through it and try to reproduce the results on your own. This exercise will be very helpful for the next few homeworks.
2. **Conditional Probability.** Suppose a couple has two children. The sample space is $S = \{bb, bg, gb, gg\}$ where we assume an equiprobable space, that is, we assume probability $1/4$ for each point. Find the probability p that both children are boys if it is known that:
 - a. At least one of the children is a boy.

$$P(bb|bb, bg, gb) = \frac{P(bb \cap bb, bg, gb)}{P(bb, bg, gb)} = \frac{P(bb)}{P(bb, bg, gb)} = \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3}$$

- b. The older child is a boy.

$$P(bb|bb, bg) = \frac{P(bb \cap bb, bg)}{P(bb, bg)} = \frac{P(bb)}{P(bb, bg)} = \frac{\frac{1}{4}}{\frac{2}{4}} = \frac{1}{2}$$

3. **Independent events.** The probability that A hits a target is $\frac{1}{4}$ and the probability that B hits the target is $\frac{2}{5}$. Both shoot at the target. Find the probability that at least one of them hits the target; that is, find the probability that A or B (or both) hit the target.

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) = P(A) + P(B) - P(A)P(B) = \frac{1}{4} + \frac{2}{5} - \frac{1}{4} \times \frac{2}{5} \\ &= \frac{5}{20} + \frac{8}{20} - \frac{2}{20} = \frac{11}{20} \end{aligned}$$

4. **Pairwise Independence vs Independence.** A fair coin is tossed twice. Assume the two tosses are independent. The sample space for this random experiment is $S = \{HH, HT, TH, TT\}$. Define three events
 - $A =$ “the first toss shows Head”
 - $B =$ “the second toss shows Head”
 - $C =$ “both tosses show the same outcome”
 - a. Calculate $P(A \cap B)$, $P(A \cap C)$, and $P(C \cap B)$. What can you say about the pairs of events (A, B) , (A, C) , and (B, C) ?

$$P(A \cap B) = P(HH, HT \cap HH, TH) = P(HH) = \frac{1}{4}$$

$$P(A \cap C) = P(HH, HT \cap HH, TT) = P(HH) = \frac{1}{4}$$

$$P(C \cap B) = P(HH, TT \cap HH, TH) = P(HH) = \frac{1}{4}$$

The pairs of events (A, B) , (A, C) , and (B, C) all describe the same outcome.

- b. Now calculate $P(A \cap B \cap C)$ and compare with $P(A) \times P(B) \times P(C)$. What does this comparison tell you?

$$P(A \cap B \cap C) = P(HH, HT \cap HH, TH \cap HH, TT) = P(HH) = \frac{1}{4}$$

$$P(A) \times P(B) \times P(C) = P(HH, HT) \times P(HH, TH) \times P(HH, TT) = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$$

This comparison demonstrates that although each of these events are pairwise independent, they are not completely independent. If both A and B are true, then C must be true, so the probability of all three events happening is not equal to the product of their probabilities.

5. **Random Variables.** Solve problems 5.86, 5.88, and 5.89 in the textbook. Full points will be given only if you show all intermediate steps clearly (this includes problems on plotting CDFs).

5.86 Let X be a continuous random variable with the following distribution:

$$f(x) = \begin{cases} \frac{1}{8} & \text{if } 0 \leq x \leq 8 \\ 0 & \text{elsewhere} \end{cases}$$

Find:

- a. $P(2 \leq X \leq 5)$

$$P(2 \leq X \leq 5) = \int_2^5 \frac{1}{8} dx = \frac{1}{8} (5 - 2) = \frac{3}{8}$$

- b. $P(3 \leq X \leq 7)$

$$P(3 \leq X \leq 7) = \int_3^7 \frac{1}{8} dx = \frac{1}{8} (7 - 3) = \frac{4}{8} = \frac{1}{2}$$

- c. $P(X \geq 6)$.

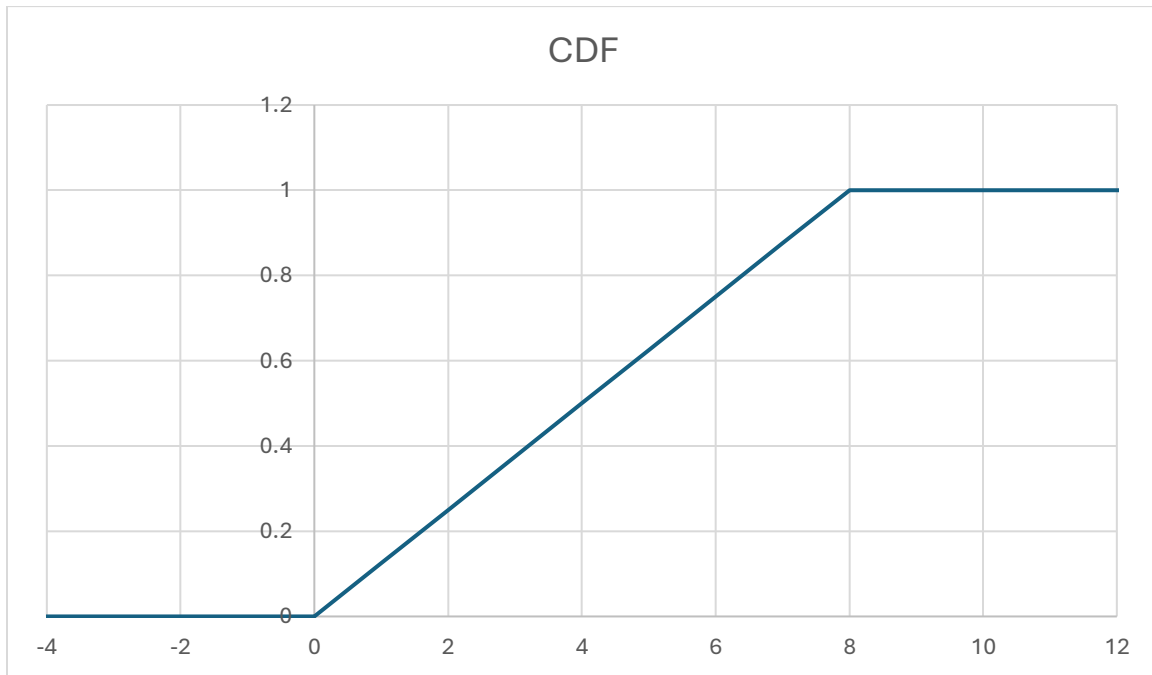
$$P(X \geq 6) = \int_6^8 \frac{1}{8} dx = \frac{1}{8} (8 - 6) = \frac{2}{8} = \frac{1}{4}$$

5.88 Determine and plot the graph of the cumulative distribution function F of the random variable X in Problem 5.86.

For $x \leq 0$, $F(x) = 0$

For $0 \leq x \leq 8$, $F(x) = \int_0^x \frac{1}{8} dt = \frac{1}{8}x$

For $x \geq 8$, $F(x) = 1$



5.89 Let X be a continuous random variable with the following distribution:

$$f(x) = \begin{cases} kx & \text{if } 0 \leq x \leq 5 \\ 0 & \text{elsewhere} \end{cases}$$

Evaluate k and find:

$$\int_0^5 kx \, dx = 1$$

$$\frac{1}{2}k5^2 = 1$$

$$k = 2/25$$

a. $P(1 \leq X \leq 3)$

$$P(1 \leq X \leq 3) = \int_1^3 \frac{2}{25}x \, dx = \frac{1}{25}(3^2) - \frac{1}{25}(1^2) = \frac{8}{25}$$

b. $P(2 \leq X \leq 4)$

$$P(2 \leq X \leq 4) = \int_2^4 \frac{2}{25}x \, dx = \frac{1}{25}(4^2) - \frac{1}{25}(2^2) = \frac{12}{25}$$

c. $P(X \leq 3)$

$$P(X \leq 3) = \int_0^3 \frac{2}{25} x \, dx = \frac{1}{25} (3^2) = \frac{9}{25}$$

6. Let S be the sample space when a pair of fair dice is tossed. Let X and Y be random variables on S , where $X(a, b) = \max(a, b)$, and Y denotes the sum of the numbers, $Y(a, b) = a + b$.

a. Find the PMF $f_X()$ of X .

Die 1	Die 2	Maximum
1	1	1
1	2	2
1	3	3
1	4	4
1	5	5
1	6	6
2	1	2
2	2	2
2	3	3
2	4	4
2	5	5
2	6	6
3	1	3
3	2	3
3	3	3
3	4	4
3	5	5
3	6	6
4	1	4
4	2	4
4	3	4
4	4	4
4	5	5
4	6	6
5	1	5
5	2	5
5	3	5
5	4	5
5	5	5
5	6	6
6	1	6

6	2	6				
6	3	6				
6	4	6				
6	5	6				
6	6	6				
x	1	2	3	4	5	6
$P(X = x)$	1/36	3/36	5/36	7/36	9/36	11/36

$$f_X(x) = P(X = x) = \begin{cases} \frac{2x-1}{36} & \text{if } x \in \{1,2,3,4,5,6\} \\ 0 & \text{otherwise} \end{cases}$$

b. Find the PMF $f_Y()$ of Y . Also, find the mean (expected) values $\mathbb{E}X$ and $\mathbb{E}Y$.

Die 1	Die 2	Sum
1	1	2
1	2	3
1	3	4
1	4	5
1	5	6
1	6	7
2	1	3
2	2	4
2	3	5
2	4	6
2	5	7
2	6	8
3	1	4
3	2	5
3	3	6
3	4	7
3	5	8
3	6	9
4	1	5
4	2	6
4	3	7
4	4	8
4	5	9
4	6	10
5	1	6

5	2	7
5	3	8
5	4	9
5	5	10
5	6	11
6	1	7
6	2	8
6	3	9
6	4	10
6	5	11
6	6	12
x	$P(Y = y)$	
2	1/36	
3	2/36	
4	3/36	
5	4/36	
6	5/36	
7	6/36	
8	5/36	
9	4/36	
10	3/36	
11	2/36	
12	1/36	

$$f_Y(y) = P(Y = y) = \begin{cases} \frac{1}{36} & \text{if } y \in \{2,12\} \\ \frac{2}{36} & \text{if } y \in \{3,11\} \\ \frac{3}{36} & \text{if } y \in \{4,10\} \\ \frac{4}{36} & \text{if } y \in \{5,9\} \\ \frac{5}{36} & \text{if } y \in \{6,8\} \\ \frac{6}{36} & \text{if } y = 7 \\ 0 & \text{otherwise.} \end{cases}$$

$$\begin{aligned} \mathbb{E}(X) &= 1\left(\frac{1}{36}\right) + 2\left(\frac{3}{36}\right) + 3\left(\frac{5}{36}\right) + 4\left(\frac{7}{36}\right) + 5\left(\frac{9}{36}\right) + 6\left(\frac{11}{36}\right) \\ &= \frac{1}{36} + \frac{6}{36} + \frac{15}{36} + \frac{28}{36} + \frac{45}{36} + \frac{66}{36} = \frac{161}{36} = 4.47\bar{2} \end{aligned}$$

$$\begin{aligned}\mathbb{E}(Y) &= 2\left(\frac{1}{36}\right) + 3\left(\frac{2}{36}\right) + 4\left(\frac{3}{36}\right) + 5\left(\frac{4}{36}\right) + 6\left(\frac{5}{36}\right) + 7\left(\frac{6}{36}\right) + 8\left(\frac{5}{36}\right) + 9\left(\frac{4}{36}\right) + 10\left(\frac{3}{36}\right) \\ &\quad + 11\left(\frac{2}{36}\right) + 12\left(\frac{1}{36}\right) \\ &= \frac{2}{36} + \frac{6}{36} + \frac{12}{36} + \frac{20}{36} + \frac{30}{36} + \frac{42}{36} + \frac{40}{36} + \frac{36}{36} + \frac{30}{36} + \frac{22}{36} + \frac{12}{36} = \frac{252}{36} = 7\end{aligned}$$

7. **Bernoulli and Binomial Distributions.** The probability that Ann hits a target at any time is $p = \frac{1}{3}$. Suppose she fires at the target 7 times. This is a Binomial experiment with $p = \frac{1}{3}, n =$

7. Find the probability that she hits the target:

- a. exactly 3 times

$$P(X = 3) = \binom{n}{k} p^k (1-p)^{n-k} = \frac{7!}{3!4!} \left(\frac{1}{3}\right)^3 \left(\frac{2}{3}\right)^4 = \frac{5040}{144} \left(\frac{1}{27}\right) \left(\frac{16}{81}\right) = \frac{80640}{314928} = \frac{560}{2187}$$

- b. at least 1 time.

$$P(X > 0) = 1 - P(X = 0) = 1 - \frac{7!}{0!7!} \left(\frac{1}{3}\right)^0 \left(\frac{2}{3}\right)^7 = 1 - \frac{128}{2187} = \frac{2059}{2187}$$

8. Solve problems 6.51, 6.52, 6.61 (a), and 6.63 (expected number E only) in the textbook.

6.51 Find $P(k)$ for the binomial distribution $B(n, p)$ where:

- a. $n = 5, p = 1/3, k = 2$

$$P(X = 2) = \binom{5}{2} \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right)^3 = \frac{80}{243}$$

- b. $n = 7, p = 1/2, k = 3$

$$P(X = 3) = \binom{7}{3} \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^4 = \frac{21}{128}$$

- c. $n = 4, p = 1/4, k = 2$

$$P(X = 2) = \binom{4}{2} \left(\frac{1}{4}\right)^2 \left(\frac{3}{4}\right)^2 = \frac{27}{128}$$

6.52 A card is drawn and replaced 3 times from an ordinary 52-card deck. Find the probability that:

- a. 2 hearts were drawn

$$P(X = 2) = \binom{3}{2} \left(\frac{1}{4}\right)^2 \left(\frac{3}{4}\right)^1 = \frac{9}{64}$$

- b. 3 hearts were drawn

$$P(X = 3) = \binom{3}{3} \left(\frac{1}{4}\right)^3 \left(\frac{3}{4}\right)^0 = \left(\frac{1}{4}\right)^3 = \frac{1}{64}$$

c. at least 1 heart was drawn

$$P(X > 0) = 1 - P(X = 0) = 1 - \binom{3}{0} \left(\frac{1}{4}\right)^0 \left(\frac{3}{4}\right)^3 = 1 - \left(\frac{3}{4}\right)^3 = 1 - \frac{27}{64} = \frac{37}{64}$$

6.61 Team B has probability $p = 0.6$ of winning each time it plays. Let X denote the number of times B wins in 4 games. Find the distribution of X .

$$P(X = 0) = \binom{4}{0} (0.6)^0 (0.4)^4 = 0.0256$$

$$P(X = 1) = \binom{4}{1} (0.6)^1 (0.4)^3 = 0.1536$$

$$P(X = 2) = \binom{4}{2} (0.6)^2 (0.4)^2 = 0.3456$$

$$P(X = 3) = \binom{4}{3} (0.6)^3 (0.4)^1 = 0.3456$$

$$P(X = 4) = \binom{4}{4} (0.6)^4 (0.4)^0 = 0.1296$$

6.63 A fair die is tossed 180 times. Find the expected number E of times the face 6 occurs.

$$E(X = 6) = np = \frac{180}{6} = 30$$

9. **The Gaussian distribution.** Read Sec. 6.4 and Example 6.5 in the textbook before answering this question. Answer questions 6.67 and 6.68 in the textbook.

6.67 Let Z be the standard normal random variable. Find

$$a. P(-0.81 \leq Z \leq 1.13) = \Phi(|-0.81|) + \Phi(1.13) = 0.2910 + 0.3708 = 0.6618$$

$$b. P(-0.23 \leq Z \leq 1.6) = \Phi(|-0.23|) + \Phi(1.6) = 0.0910 + 0.4452 = 0.5362$$

$$c. P(0.53 \leq Z \leq 2.03) = \Phi(2.03) - \Phi(0.53) = 0.4788 - 0.2019 = 0.2769$$

$$d. P(0.15 \leq Z \leq 1.50) = \Phi(1.50) - \Phi(0.15) = 0.4332 - 0.0596 = 0.3736$$

6.68 Let Z be the standard normal random variable. Find

$$a. P(Z \leq 0.73) = 0.5 + \Phi(0.73) = 0.5 + 0.2673 = 0.7673$$

$$b. P(Z \leq 1.8) = 0.5 + \Phi(1.8) = 0.5 + 0.4641 = 0.9641$$

$$c. P(Z \geq 0.2) = 0.5 - \Phi(0.2) = 0.5 - 0.0793 = 0.4217$$

$$d. P(Z \geq -1.5) = 0.5 + \Phi(|-1.5|) = 0.5 + 0.4332 = 0.9332$$

$$e. P(Z = 1.8) = 0$$

$$f. P(|Z| \leq 0.25) = 2 \times \Phi(0.25) = 2 \times 0.0987 = 0.1974$$

10. **SciPy stats module.** The stats module in SciPy allows users to sample from many continuous distributions. [A full list can be found here](#). We've already seen the normal (Gaussian) distribution in class. How will you sample a set of $N = 10,000$ random numbers in SciPy from (i) the normal distribution, (ii) the exponential distribution, and (iii) the Poisson distribution?

- a. Give the expression for the probability density/mass functions for each of these three distributions.

Normal distribution: $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

Exponential distribution: $f(x, \lambda) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$

Poisson distribution: $P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$

- b. Using the N samples, plot the histograms of each of the three distributions.

